Sorting-1

Agenda:

1. Sorting Basics
2. Min Cost to remove elements
3. Find K'th Max Element via Selection Sort
4. Insertion Sort
5. Stable Sorting & In-Place Sorting

Built-in methods

- Arrays: Arrays.sort(A)
- ArrayList: Collections.sort(A) — Both return void
- T.C = $O(N \log N)$
  S.C = $O(N)$

Minimum Cost To Remove Elements

```
Public class MinCostRemoval {
    Public static int solve (Integer [] A) {
        Arrays.sort(A);          // sort in ascending
    int i=0, j = A.length-1      // Reverse the array
    while (i<j) {
        int temp = A[i];
            A[i] = A[j];
            A[j] = temp;
            i++;
            j--;
        }
    int totalCost = 0
      for (int k = 0; k < A.length; k++) {
            totalCost += (k+1) * A[k]
      }
        return totalCost;
    }
}
```

Find Kth max using Selection Sort

- Repeatedly finding the max/min element from the unsorted part and putting it at the end (or beginning).
- Finding Max: Iterate through the array; if A[i] > A[idx], update the max idx.
- Complexity: $O(K \times N)$ Sorting whole array → T.C = $O(N^2)$
  S.C = $O(1)$

Kth Max (Selection Sort (Descending order))

```
public class KthMaximum(int[]arr, int k){
    int n = arr.length;
    for(int i=0 ; i < k ; i++ ){
        int maxIndex=i;
        for (int j =i+1 ; j< n ; j++){
            if ( arr[j] > arr[maxIndex] ){
                maxIndex =j;
            }
        }
        int temp = arr[i]
        arr[i] = arr [maxIndex]
        arr [maxIndex] = temp
    }
    return arr[k-1] ;
}
```

→ assume position i as largest

if current element greater than assumed max

Swap index of largest element to front

Dry Run → Array : [7, 10, 4, 3, 20, 15], K = 3

Pass		Array State
1	→	20, 10, 4, 3, 7, 15
2	→	20, 15, 4, 3, 7, 10
3	→	20, 15, 10, 3, 7, 4

3rd max = 10

Insertion Sort → Use : Sorting "a running stream" of data

↳ take a new element and insert it into its correct position among the already sorted elements by shifting others.

Complexity : TC : $O(N^2)$ for whole arr, $O(N)$ to insert single element into sorted list

SC : $O(1)$